MIL-R-50090D(MU)
19 August 1970
SUPERSEDING
MIL-C-50090C(MU)
7 October 1969

MILITARY SPECIFICATION

RIOT CONTROL AGENT, CS1

1. SCOPE

1.1 This specification covers one type of riot control agent, hereinafter referred to as CS1.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Steel Strapping, Flat.
TT-W-572 - Wood-Preservative, Water-Repellent.
UU-S-48 - Sacks, Shipping, Paper.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-D-729 - Drums: Metal, 55-Gallon (For Shipment of Noncorrosive Material).
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-117 - Bags, Interior Packaging.
MIL-B-131 - Barrier Material, Water Vaporproof, Flexible, Heat-Sealable.
MIL-S-12613 - Silica, Colloidal, Synthetic.
MIL-T-22085 - Tape, Pressure-Sensitive Adhesive, Preservation and Sealing.
MIL-C-51029 - Chemical Agent, CS.

FSC 1365

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND (EDGEWOOD ARSENAL)

C18-60-1283 - Marking of Packing for Chemical Agent, CS-1 or CS-2.
E18-60-1294 - Marking, Packing; Metal Shipping Container for Riot Control Agent CS, CS1, or CS2.
D18-60-1322 - Palletization for Riot Control Agent (CS-1 or CS-2).
D150-2-2 - Color Coding for Chemical Munitions.

PUBLICATIONS

EDGEWOOD ARSENAL PURCHASE DESCRIPTION

196-21-87 - Bottle, Screw-Cap, Plastic, 3-1/2-Gallon.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

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3. REQUIREMENTS

3.1 Chemical characteristics. CS1 shall be a homogeneous mixture of 95.0 ± 1.0 percent by weight Chemical Agent, CS conforming to MIL-C-51029 and 5.0 ± 1.0 percent by weight silica aerogel conforming to MIL-S-12613 when tested as specified in 4.2.4.1.

3.2 Ultimate particle size. CS1 shall have an ultimate particle size no greater than 9 microns at 50-percent by weight cumulative and no greater than 12 microns at 80-percent by weight cumulative when tested as specified in 4.2.4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the CS1 offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination for preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105 using an AQL of 1.0 percent defective.

4.2.2.2 For specimen test. Sampling for specimen test shall be conducted in accordance with table I. A representative specimen of no less than 1 pound (lb) shall be removed from each sample packing container and placed in a suitable clean, dry, closed container labeled to identify the lot and container from which it was taken (see 6.5).

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Table I. Sampling for test

Number of packing containers in lot or batch	Number of sample packing containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For leakage test. Sampling for leakage test shall be conducted in accordance with MIL-STD-105, level S-4 using an AQL of 1.0 percent defective.

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined in accordance with MIL-STD-105 and the following classification of defects:

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Contents per container not as specified
102	Container not as specified
103	Container damaged (punctured or torn)
104	Container leaking
105	Container closure not as specified
106	Outside of container contaminated with CSl
107	Marking missing, illegible, or incorrect
108	Palletization incorrect (where required)

4.2.3.2 For specimen test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4.1 and 4.2.4.2. Failure of either test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For leakage test. The sample unit shall be one filled and closed plastic bottle or one filled and sealed plastic bag, as applicabl Each sample bottle or bag taken as specified in 4.2.2.3 shall be tested as specified in 4.2.4.3.

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4.2.4 Tests. Tests shall be conducted as follows (see 6.5):

4.2.4.1 Composition. Weigh approximately 10 grams (g) of the specimen to the nearest milligram (mg). Transfer to a tared crucible. Ignite the crucible and contents at 700° ± 10° C until a constant weight is achieved. Cool in a desiccator and reweigh. Calculate the percent silica aerogel present as follows:

$$\text{Percent silica aerogel} = \frac{100A}{B}$$

where: A = Weight of the residue in grams, and
B = Weight of the specimen in grams.

Calculate the percent CS as follows:

$$\text{Percent CS} = 100 - C$$

where: C = Percent silica aerogel calculated above.

4.2.4.2 Ultimate particle size.

(a) Apparatus. Use a microscope with the following equipment:

(1) Ten power compensating eyepiece.

(2) Forty-three power (4 millimeter) apochromatic objective with a 0.95 numerical aperture and a correction collar for cover glass thickness variation.

(3) Aplanatic apochromatic sub-stage condenser of high numerical aperture.

(4) Stage micrometer, consisting of a reticle with a millimeter scale divided into 100 divisions, mounted on a standard 1-inch by 3-inch glass slide.

(5) Projection equipment consisting of a right-angled eyepiece lens and a projection screen or a ground glass screen. The distance between the microscope and the screen must be capable of being varied.

(b) Calibration. Place the micrometer scale slide on the stage of the microscope in normal position. Project the image onto the screen, adjusting the distance between the microscope and the screen

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until 67 divisions of a millimeter-graduated transparent ruler equal 10 divisions of the image on the screen. One division of the ruler will now equal 1.5 microns.

(c) Slide preparation. Weigh 0.100 g of specimen into each of two 200-milliliter (ml), tall-form beakers. Add 20 ml of light mineral oil to the first beaker and mix for 5 ± 1/2 minute with an electric stirrer having twin blades totaling no more than 1.7 inches in diameter and operating at a speed of 1650 ± 150 revolutions per minute. While continuing to stir the mixture, withdraw a portion of the mixture with an eyedropper, taking it from a point just above the stirrer blades and approximately one-fourth the diameter of the beaker away from the inside wall. Immediately wipe the outside of the eyedropper. Place one drop of the mixture on a clean, dry 1-inch by 3-inch microscope slide and cover with a No. 1, 3/4-inch diameter cover glass. In the same manner, prepare a second slide from the first beaker. Then, following the same procedure with the second beaker, prepare a second pair of slides. Label the four slides to identify the beakers and specimen represented.

(d) Measurements. Mount the first slide on the microscope stage and project the image onto the screen. Place the transparent ruler horizontally across the image so that the field is approximately bisected. Measure to the nearest millimeter the maximum length of the horizontal projection of each particle successively encountered. Disregard the orientation of the particles and exclude clusters or agglomerates. When the edge of the field of vision is reached, continue measuring particles parallel to those already measured but sufficiently distant from them so that none of the previous particles are re-measured. As the particles are measured, tabulate them into class intervals as illustrated in table II. Measure the particles until no less than 200 particles having a length over 1.5 microns are counted. A hand magnifying glass may be used to aid the measurement. If one-fourth or more of the field of vision is occupied by clusters or agglomerates, discard both slides from the beaker represented, disregard the results from these slides, and prepare two replacement slides from a new beaker. Photomicrograph techniques or direct size measurement by means of an eyepiece calibrated grating may be used in lieu of screen projection for particle counting.

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Table II. Frequency distribution

Class interval in millimeters*	Midpoint of class in microns*	Number of particles in class (Data shown are examples)
0.0 to 0.5	----	Disregard
0.6 to 1.5	1.5	87
1.6 to 2.5	3.0	115
2.6 to 3.5	4.5	59
3.6 to 4.5	6.0	26
4.6 to 5.5	7.5	14
5.6 to 6.5	9.0	6
6.6 to 7.5	10.5	1
7.6 to 8.5	12.0	1
8.6 to 9.5	13.5	0
9.6 to 10.5	15.0	0
10.6 to 11.5	16.5	0
11.6 to 12.5	18.0	0

*Invariable data

(e) Calculation of ultimate particle size. Prepare a frequency distribution similar to table II for each of the four slides. After recording the total number of particles in each micron class, prepare a tabulation as shown in table III (which is a sample recording of typical data) for each slide. Use the following formula to calculate the percent by weight cumulative:

$$\text{Percent by weight cumulative} = \frac{nd^3 \text{ cumulative} \times 100}{\text{total } nd^3}$$

Using logarithmic graph paper, plot the data from each slide separately with the class size in microns plotted along the logarithmic scale and the percent by weight cumulative plotted along the nonlogarithmic scale. From the curve produced, determine the particle size, in microns, at the 50-percent by weight cumulative intercept and at the 80-percent by weight cumulative intercept. Average the four results.

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Table III. Sample data

d	n	d^3	nd^3	nd^3 cumulative	Percent by weight cumulative
1.5	87	3.375	294	294	1.1
3.0	115	27.00	3,105	3,399	12.3
4.5	59	91.13	5,377	8,776	31.8
6.0	26	216.0	5,616	14,392	52.2
7.5	14	421.9	5,907	20,299	73.7
9.0	6	729.0	4,374	24,673	89.5
10.5	1	1,158.0	1,158	25,831	93.7
12.0	1	1,728.0	1,728	27,559	100.0
13.5	---	2,460	-----	------	-----
15.0	---	3,375	-----	------	-----
16.5	---	4,492	-----	------	-----
18.0	---	5,832	-----	------	-----

d = midpoint of class in microns.
n = number of particles tallied in micron class.

A plot of the sample data shows that the particle size equals 5.5 microns at 50-percent by weight cumulative and 8.3 microns at 80-percent by weight cumulative.

4.2.4.3 Leakage of plastic bottles and bags. Conduct the test in a well-ventilated hood. Lift the sample plastic bottle or bag until the bottom of the bottle or bag is 24 ± 3 inches above a horizontal surface. Hold the bottle in the normal upright position with the cap at the top. Hold the bag so that the heat-sealed edge is the highest part of the bag. Drop the bottle or bag onto the horizontal surface. There shall be no visible emission of CS1 from the bottle or bag during or after impact.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A in 8-lb bags or 8-lb plastic containers as specified (see 6.2).

5.1.1 Eight-lb bags. Eight lb ± 2 ounces (oz) of CS1 shall be packaged in a 17-1/2-inch by 25-inch or a 16-inch by 24-inch bag conforming to type I, class E, style 1 of MIL-B-117. The bag shall be made of heat-sealable, flexible material conforming to class 1 of MIL-B-131. The bag shall be closed by heat sealing and shall then pass the leakage test specified in 4.2.4.3. The bag shall then be inserted into a snug-fitting sack conforming to style A, type II, construction No. 1 of UU-S-48. The sack shall be stitched closed as specified in UU-S-48, without stitching or otherwise damaging the inner heat-sealed bag. The outer surfaces of the bag and the sack shall be free of CS1 contamination.

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5.1.2 Eight-lb plastic bottles. Eight lb ± 2 oz of CS1 shall be packaged in a plastic bottle conforming to Edgewood Arsenal Purchase Description 196-21-87 and in accordance with any additional requirements specified in the contract or order (see 6.2). The filled bottle shall be closed by tightening a properly engaged screw cap conforming to the screw cap requirements of Edgewood Arsenal Purchase Description 196-21-87 onto the threaded neck of the bottle to a torque of 40 ± 5 inch-pounds. Before applying tape to the closure, the filled and closed bottle shall pass the leakage test specified in 4.2.4.3. The tightened screw cap closure shall be further secured to the neck of the bottle with tape of 2-inch minimum width conforming to type II of MIL-T-22085. The tape shall be tightly wound around the neck of the bottle and the cap for a minimum of two complete turns. The tape shall extend from the top of the cap to a circumferential line around the bottle spout no less than 3/4 inch below the bottom of the tightened cap. The plastic bottle and closure shall be free of CS1 contamination.

5.2 Packing. Packing shall be level A or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Eight-lb bags. Ten bags of CS1, packaged in sacks as specified in 5.1.1, shall be packed in a 55-gallon fully-removable head drum conforming to Department of Transportation (DOT) Specification 5B or 6C. The drum shall have an inside diameter of 22-1/2 ± 1/16 inch, as measured between opposite points 1/2 inch below the top edge of the drum, and shall have an interior height of no less than 34-3/8 inches, as measured on the vertical axis of the drum between the bottom head and a diameter across the top of the upper curl. The drum shall be cleaned and painted as specified in PPP-D-729. The interior arrangement shall consist of two tiers of five sacks per tier. The tiers shall be separated by fiberboard pads conforming to grade W5c of PPP-F-320. Additional blocking in the form of fiberboard cells of the same material shall be used, where necessary, to assure a tight pack. The filled drum shall be closed and the cover tightly secured thereto by means of the bolted locking ring.

5.2.1.2 Eight-lb plastic bottles. Five plastic bottles of CS1, packaged as specified in 5.1.2, shall be packed side by side, in an upright position, with pouring spouts facing in alternate directions in a close-fitting fiberboard box. The fiberboard box shall be enclosed in a heat-sealed bag and placed in a nailed wood box in accordance with submethod IA-14 of MIL-P-116. The fiberboard box shall conform to style RSC, type CF, class domestic, variety SW, grade 125 of PPP-B-636. The inside dimensions of the fiberboard box shall be 32 inches by 11 inches by 17-1/2 inches. Fiberboard separators conforming to type CF, class domestic, variety SW, grade optional of PPP-F-320 shall completely separate the plastic bottles and assure a tight pack. A

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flat fiberboard pad, extending the full length of the box shall be placed above the handle and between the spouts. The pad shall extend 1/4 ± 1/8 inch above the closure of the spout. The fiberboard for the pad shall be of the same material as that specified for the separators. The fiberboard box shall be closed and sealed in accordance with the requirements in the appendix to PPP-B-636. The heat-sealed bag enclosing the fiberboard box shall be formed from barrier material conforming to class 1 of MIL-B-131 and shall have inside dimensions of 31 inches by 44-3/4 inches. The nailed wood box enclosing the fiberboard box and heat-sealed bag shall conform to class 2, style 4-1/2, for a type 3 load exceeding 100 lb but not exceeding 250 lb of PPP-B-621. The inside dimensions of the wood box shall be 33-1/4 inches by 11-3/4 inches by 18-1/2 inches. Fiberboard pads conforming to grade W5c or W6c of PPP-F-320 shall be inserted between the heat-sealed bag and all six interior surfaces of the wood box. The wood box shall be closed and strapped with galvanized strapping conforming to type I, class B of QQ-S-781. The empty box or the finished wood parts thereof shall be completely immersed for no less than 3 minutes in a water-repellent wood-preservative conforming to composition A of TT-W-572.

5.2.2 Level C. Unless otherwise specified (see 6.2), 125 ± 1 lb of CS1 shall be packed in a heat-sealable bag placed within a 55-gallon fully-removable head drum conforming to DOT specification 5B. The bag shall be formed from material having a thickness of no less than 0.005 inch and conforming to type I, class 1, grade B or C, finish 1 of L-P-378. All seams of the bag, including the closure seam, shall be completely heat sealed. The filled drum shall be closed and the cover tightly secured thereto by means of the bolted locking ring. Care shall be taken so that no CS1 is deposited between the drum and the bag.

5.3 Marking.

5.3.1 Eight-lb bags. Each 8-lb bag shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with Drawing E18-60-1294.

5.3.2 Eight-lb plastic bottles. Each 8-lb plastic bottle shall be marked in accordance with MIL-STD-129. In addition, the centers of the diagonally opposite vertical edges to the right of the identification marking shall be color coded with one red stripe on a gray background and marked with the identification symbol "CS1" in accordance with Drawing D150-2-2. Shipping containers shall be marked in accordance with Drawing C18-60-1283.

5.3.3 Level C drums. Drums shall be marked in accordance with Drawing E18-60-1294.

5.4 Palletization. When specified in the contract or order (see 6.2), shipping containers of CS1, packed as specified in 5.2.1.2, shall be palletized in accordance with Drawing D18-60-1322.

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6. NOTES

6.1 Intended use. CS1 is intended for use in grenades and riot control agent dispersers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Unit container required (see 5.1).
(c) Additional requirements for plastic bottle, if required (see 5.1.2).
(d) Level of packing required (see 5.2).
(e) Quantity required for level C pack (see 5.2.2).
(f) Palletization, if required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Caution. Tests should be performed under a hood. Attention is called to Army Munitions Command Regulation 385-237, "Safety Guide for Processing, Filling, Handling and Decontamination of CS and CS1."

6.6 International interest. Certain provisions of this specification are the subject of international standarization agreement (see Serial No. 7, Supplement B to Quadripartite Standardization List Sheet Report). When amendment, revision or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take the appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodian:

Army - MU

International interest (see section 6)

Preparing activity:

Army - MU(EA)

Project No. 1365-A047

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POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA–QAES
Edgewood Arsenal, Maryland 21010

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 64 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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